

Empirical Study on the Impact of Stock Splits in the Indian Textile Industry

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ABSTRACT

Due to ease of access of using modern technology, the numbers of investors in the stock market have been rising exponentially. The investors having a profit motive, often, become victims to window-dressing of investments. Stock splits are generally perceived by the investors to be fruitful investment opportunities. In fact, Greenbatts, Masulis, and Titman, (1984) found that a stock split is one of the ways to signal the future growth of the company. However, in recent times with the large number of options and the rapidly progressing markets, a stock split does not necessarily imply great future prospects for a company. With many recent studies focusing on the change in liquidity of the companies after a split, this paper looks at the post-split stock of a company from the investor's perspective. This paper studies data to find the truth behind common perceptions regarding a stock split. The textile sector, one of the most important sectors in India has been taken into consideration while testing the hypotheses. It takes into account various factors, particularly the total turnover of the stocks post-split and the average returns in the short run as well as in the long run.

INTRODUCTION

The Bombay Stock Exchange, with the largest number of listed companies in the world, is also one of the most important. Daily, millions of transactions occur and the BSE SENSEX has adverse effects on almost all other sectors from agriculture to politics. The stock prices of companies and commodities are interdependent. Say for example, a rise in crude prices leads to a fall in the stock prices of companies heavily dependant on fuel based transportation. Hence, any major change in one listed company tends to have a domino effect on the rest.

The Investors in the stock market need a keen eye and a good amount of knowledge to profit from the fluctuating stock prices. Wise investments can lead to gains much higher than those possible from other investment avenues. There are several factors that affect the fluctuations in the price of a stock. Major decisions of a company greatly impact its stock prices. This paper investigates the effect of a stock split on the stock prices, keeping in mind average returns to the investors.

A stock split can be simply defined as a corporate action in which a company divides its existing shares into multiple shares. Although the number of shares outstanding increases by a specific multiple, the total rupee value of the shares remains the same compared to pre-split amounts, because the split does not add any value. With the main objective of making the stock more attractive to a wider range of investors at reduced prices, corporates often look at stock splits as a solution to the problem of very expensive stocks.

Hence, the direct effects of the stock split include the increase in the number of shares floating in the market, the decrease in the Face Value of a stock, and with the decrease in the market value of the share expected to be directly proportional to the reduction in the Face Value.

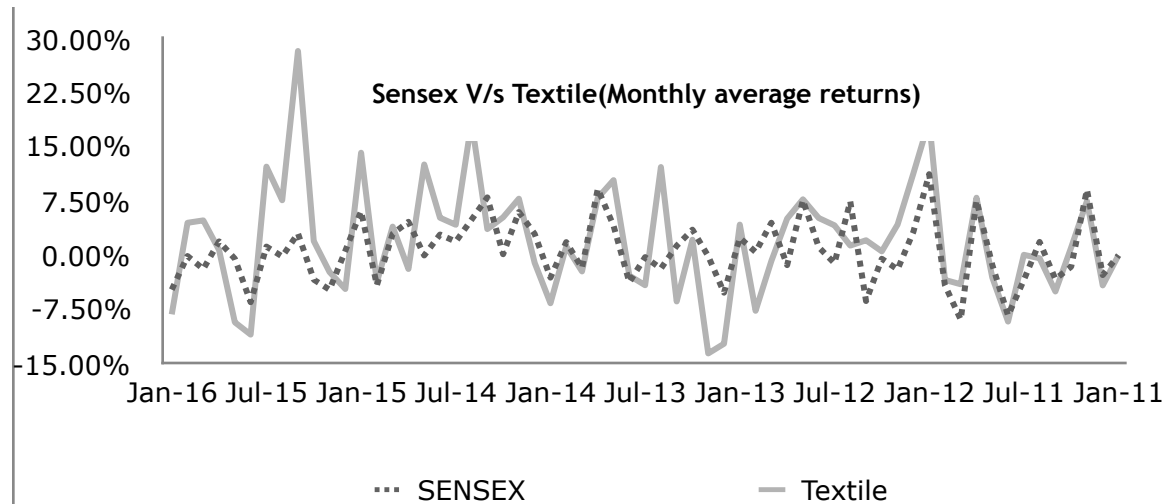
However, the effects of a stock split do not end there. There are a large number of indirect implications of a company undertaking a stock split. A split plays into the public sentiment of "more is better", people prefer to have more number of shares at a lower rate rather than a fewer higher valued shares. Another area this paper aims to address is the common perception of a stock split unleashing the underlying growth of the company, which translates to an expected increase in returns for the investors after the split.

LITERATURE REVIEW

Muscarella and Vetsuypens (1996), show that liquidity after stock split improves and results in wealth gains for investors. Menendez and Anson, (2003) point out that in addition to signaling, firms that go for a split focus on improved liquidity and an optimal trading range. Mayank Joshipura obtained a result proving that there is a significant improvement seen in liquidity surrounding announcement and effective day of stock split. Conversely, Gupta & Gupta, (2007) found significant abnormal returns surrounding ex-split day and not surrounding announcement day. Saraswathi Thirunellai, (2013) says that the historical price movement of a split stock indicates that for a given holding period (5-day, 10-day, 30-day, 60 day and 360 day), buyers of stock in the post-split period are inadequately rewarded compared to the pre-split/pre-announcement buyers. Baker et al (2009) argue that stock splits provide a good laboratory for studying managers' behavior because splits "are not associated with any confounding, 'real' motivation involving firm fundamentals".

SCOPE OF STUDY

For the study, data from the textile sector of India has been used. The textile sector is the second largest employer in India after the agricultural sector and has a large impact on the country's GDP. Topping the table in the number of stock splits in recent years, the Textile sector is an ideal sector to compare and find the extent of effects of the stock split. The Graph 1 below shows that the BSE SENSEX and the Textile Sector move similarly over 5 years. The y-axis indicates the average percentage change in the values of the stocks in the Textile Sector and the percentage change in S&P BSE SENSEX respectively. The x-axis indicates the time period (January, 2011 to January, 2016). We can infer from Graph 1 give below that the stocks in the textile sector and the SENSEX have a positive correlation over a long period of time.



Graph 1

RESEARCH METHODOLOGY

The research was quantitative in nature and data was collected from reliable secondary sources. The following data was collected from January, 2011 to April, 2016 on a yearly, monthly and daily basis respectively

- BSE SENSEX.
- Financial statements and historical share prices of leading companies in the Textile Sector: Lambodhara Textiles, Welspun India, Risa International, Samtex Fashions, Tuni Textile Mills, Century Textiles, Orbit Exports, Alok Industries, Seasons Textiles, Bombay Rayon.

Data analysis was done using Descriptive Statistics (Mean, median, mode, standard deviation, percentages, range, significance of difference etc.). Hypotheses tests were conducted using Analysis of Variance – One Way, Analysis of Variance – Two Way; with replication, and Chi-Square Testing.

RESEARCH OBJECTIVE

The objective of this paper is to obtain the factual effects of a stock split instead of accepting the common perceptions of investors. The objective of the paper is to study the following:

1. Impact of the stock-split on the total turnover in the short-run, 30days pre-split and post-split.
2. Long-term impact of the stock-split on the average returns offered
3. A comparison on the average returns for split versus non-split stocks.

PROPOSED HYPOTHESIS

Hypothesis - H1

- H₁₀: There is no significant difference in the total turnover of a stock one month before the split and one month post-split
- H_{1a} There is a significant difference in the total turnover of a stock one month before split and one month post-split

Hypothesis - H2

- H₂₀ Average monthly returns of a stock are independent of the split status within a period of 9 months after the split
- H_{2a} Average monthly returns of a stock are not independent of the split status within a period of 9 months after the split

Hypothesis - H3

- H₃₀ The average returns of the non-split stock and the average returns of a split stock in the textile industry in India are independent of each other
- H_{3a} Average returns of the stocks are not independent of the split status in the textile industry

TESTING OF HYPOTHESES

Hypothesis 1

- H_{1a0}: There is no significant difference in the total turnover of the stock 1 month before split and the total turnover of the stock 1 month post-split.
- H_{1aa}: There is significant difference in the total turnover of the stock 1 month before split and the total turnover of the stock 1 month post-split.
- H_{1b0}: There is no significant difference in the total turnover for the months September-October, 2015 as compared to the total turnover for the months October-November, 2015
- H_{1ba}: There is significant difference in the total turnover for the months September-October, 2015 as compared to the total turnover for the months October-November, 2015

Explanation:

To test this hypothesis, two scenarios were considered. The testing was done on a split-stock one month pre-split and one month post-split. The same time period was used to test the non-split stocks.

The six stocks studied were: Lambodhara Textiles (Split Stock), Orbit Exports (Non-Split Stock), Risa International (Split Stock), Century Textiles (Non-Split Stock), Samtex Fashions (Split Stock) and Bombay Rayon (Non-Split Stock)

In the first scenario, stocks that had undergone split were studied individually to determine whether there is significant difference in the total turnover.

Scenario 1: The Split stock under consideration here is Lambodhara Textiles. Refer to H_{1a0} and H_{1aa}

The results of the ANOVA Test have been given in Table 1 and Table 2 below

Alpha Value: 0.05

Table 1; Summary:

<i>Groups</i>	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>
After Split	20	185797923	9289896.15	99769289188060
Before Split	20	495491479	24774573.95	494933767106818

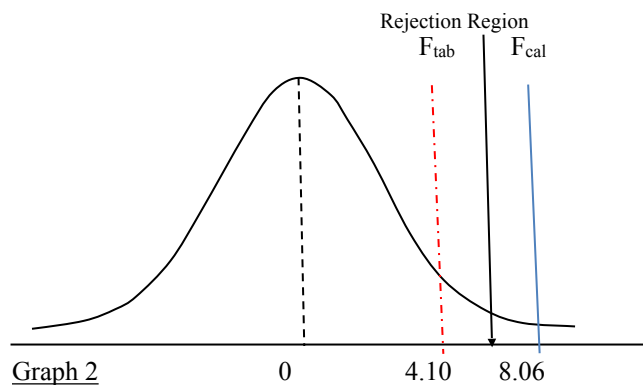
Table 2; ANOVA:

Source of Variation	SS	Df	MS	F	P-value	F crit
Between Groups	2397752465698130	1	2397752465698130	8.06	0.01	4.10
Within Groups	11299358069602700	38	297351528147439			
Total	13697110535300800	39				

Result: Since P-Value is less than Alpha, reject H_{1ao}

Since $F_{cal} > F_{tab}$ we reject H_{1ao} at 5% Level of Significance. (Refer to Graph 2)

Conclusion for H_{1a} : There is a significant difference between the total turnover of Lambodhara Textiles one month before split and one month after split.



Note: On repeating the above test multiple number of times on companies that have undergone a stock split, the following conclusion has been made. There exists a significant difference in the total turnover of a company one month after split as compared to one month before the split.

In the second scenario, stocks that had not undergone split were studied individually to determine whether there is significant difference in the total turnover.

Scenario 2: For the same time period as above, we test the data of a non-split stock Orbit Exports. Refer to H_{1b0} and H_{1ba} . The results of the ANOVA Test have been given in Table 3 and Table 4.

Alpha Value: 0.05

Table 3; Summary:

Groups	Count	Sum	Average	Variance
Oct-Nov, 2015	20	16726597	836329.85	2862402610668.45
Sept-Oct, 2015	20	12675982	633799.1	256499961589.57

Table 4; ANOVA:

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	410187046956	1	410187046956	0.26	0.61	4.10

Within Groups 59259148872902 38 1559451286129

Total 59669335919858 39

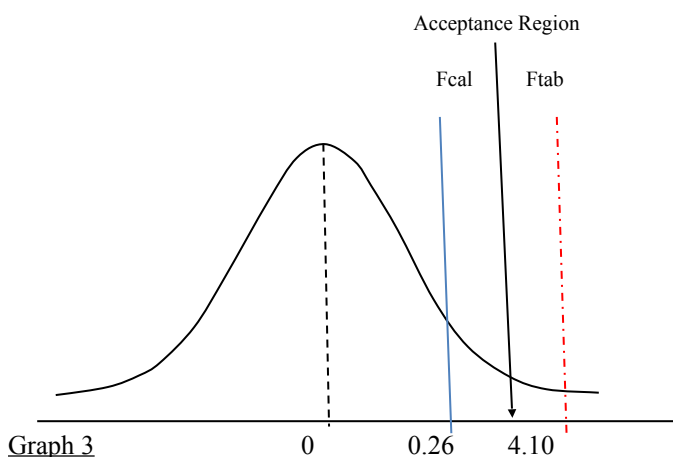
Result: Since P-Value is greater than Alpha, fail to reject H_{1b0}

Since $F_{cal} < F_{tab}$, fail to reject H_{1b0} at 5% level of significance. (Refer to Graph 3)

Conclusion for H_{1b} : There is no significant difference in the total turnover of Orbit Exports in the months Sept-Oct as compared to Oct-Nov.

Note: On repeating the above test multiple number of times, the following conclusion has been made. There is no significant difference obtained in the total turnover of a company that did not undergo a stock split in the same time period as that studied for a company that did undergo a stock split.

Conclusion for H_1 : For split stocks, there is a significant difference in the total turnover one month pre-split and one month post-split. For the same time periods, there is no significant difference in the total turnover.



Hypothesis 2

H_{20} : Average monthly returns of a stock are independent of the split status within a period of 9 months after the split.

H_{2a} : Average monthly returns of a stock are not independent of the split status within a period of 9 months after the split.

Note: The data under study is the monthly average returns of the stocks over a period of 9 months.

Table 5; Chi-Square Test to test H_2 :

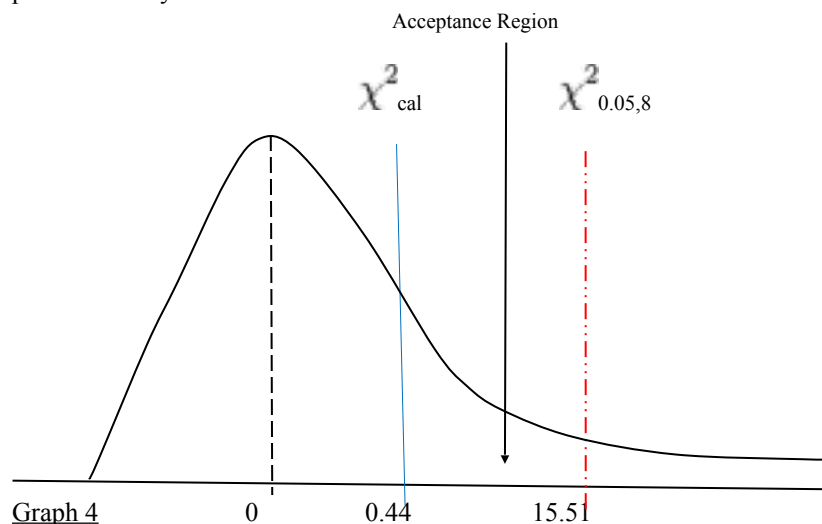
	Observed(O) Monthly Returns			Expected (E) Monthly Returns		(O - E) ^2 / E		
Date	Split Stocks	Non Split Stocks	Totals	Split Stocks	Non Split Stocks	Split Stocks	Non Split Stocks	Totals
1-Apr-16	-0.02	0.09	0.07	0.05	0.02	0.08	0.16	0.25
1-Mar-16	0.05	0.07	0.11	0.08	0.04	0.01	0.02	0.03

1-Feb-16	-0.18	-0.21	-0.4	-0.26	-0.13	-0.02	-0.05	-0.07
1-Jan-16	-0.09	-0.08	-0.17	-0.11	-0.06	-0.01	-0.01	-0.02
1-Dec-15	0.03	-0.02	0.01	0.01	0	0.05	0.1	0.16
1-Nov-15	0.1	0.07	0.17	0.11	0.06	0	0	0
1-Oct-15	0.03	0.13	0.16	0.11	0.05	0.05	0.1	0.15
1-Sep-15	0	-0.02	-0.02	-0.02	-0.01	-0.02	-0.03	-0.05
1-Aug-15	-0.16	-0.14	-0.3	-0.2	-0.1	-0.01	-0.01	-0.02
Totals	-0.24	-0.12	-0.36	-0.23	-0.13	0.15	0.29	0.44

Acceptance Region: 95%
 Rejection Region: 5%
 Degrees of Freedom: 8
 Chi-Square Calculated: 0.44
 P Value: 0.9996
 Critical Value: 15.51

Result: Fail to Reject H_{20} at 5% level of significance. (Refer to Graph 4)

Conclusion: The average growth rate is independent of Split Status of the stocks over the latest 9 months during the period of study.



Hypothesis 3

H_{30} : The average returns of the non-split stock and the average returns of a split stock in the textile industry in India are independent of each other.

H_{3a} : Average returns of the stocks are not independent of the split status in the textile industry in India.

Note: The data used in the testing below is the average returns of the each stock over the last five years.

Table 6; Chi-Square Test to test H3

	Observed (O) Yearly Returns			Expected (E) Yearly Returns		(O - E)^2 / E		
Date	S p l i t Stocks	N o n S p l i t Stocks	Totals	S p l i t Stocks	N o n S p l i t Stocks	S p l i t Stocks	Non Split Stocks	Totals
Apr'15-Apr'16	0.25	-0.09	0.16	0.12	0.04	0.13	0.41	0.54
Apr'14-Apr'15	2.1	0.89	2.99	2.28	0.7	0.01443	0.0467	0.06113
Apr'13 to Apr'14	0.8	0.23	1.03	0.79	0.24	0.0002	0.00065	0.00085
Apr'12-Apr'13	0.09	0.02	0.11	0.08	0.03	0.00035	0.00112	0.00147
Apr'11-Apr'12	-0.2	-0.11	-0.31	-0.24	-0.08	-0.00612	-0.01981	-0.02593
Totals	3.04	0.94		3.03	0.93	0.13886	0.43866	0.57

Acceptance Region: 95%

Rejection Region: 5%

Degrees of Freedom: 4

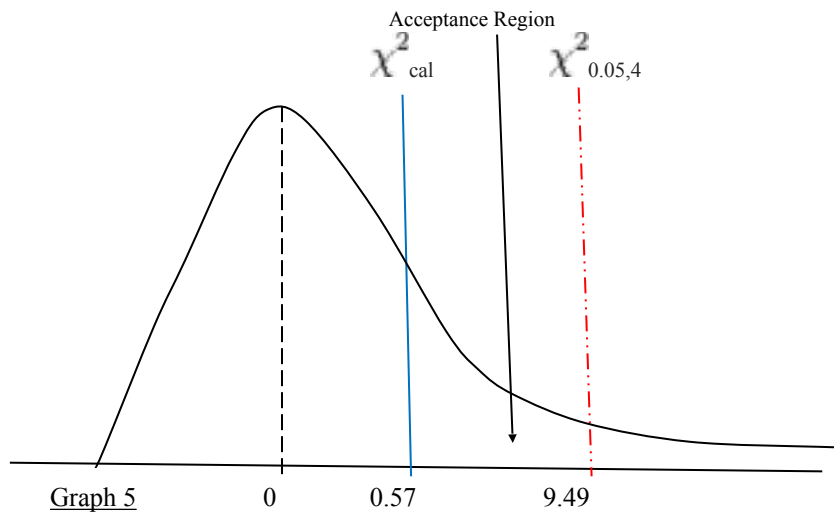
Chi-Square Calculated: 0.57

P Value: 0.9612

Critical Value: 9.49

Result: Fail to reject H_{30} at 5% level of significance. (Refer to Graph 5)

Conclusion: There is no significant difference between the average returns rate of the split and the non-split stocks over the last 5 years.



OBSERVATION AND REMARKS

Based on the results obtained above, there are the following can be inferred.

The first hypothesis shows that there exists a significance difference in the total turnover of a company just after a split for a period of one month. This significant difference can be attributed to the market sentiments, where investors buy into the euphoria of the split announcement, expecting windfall gains. Most of the initial excitement withers down within a month. The study indicates that in the long-run, stock-split doesn't really impact the total turnover. The rationale behind this behavior can be explained by investors looking for good news in the market.

Average monthly returns analysis reveals that there is no significant difference between the average monthly returns of split versus non-split stock. Thus, the study can conclude that average returns of a stock are not impacted by the strategy of splitting the stock. This further re enforces that stock splitting only achieves the objective of increasing the liquidity of the stock and it does not impact the returns of the stock.

There is no significant difference between the average growth rate of the split and the non-split stocks over the last 5 years. This shows that investors and company management should not expect significant change in annual returns through the strategy of splitting a stock. We can infer that Stock splitting only serves the purpose of increasing the liquidity of the stock. It might be worthwhile to explore whether stock splits impact average returns in the short run which is less than 6 months.

From the data under study, we have observed the average annual returns of the stocks which have split are superior to those that have not split. This can be attributed to their stronger fundamentals. Thus, one can conclude that only stocks with strong fundamentals would go for a split. So, from an investor perspective, a split stock can be an indicator of strong fundamentals and hence, possibly a good investment.

LIMITATIONS OF STUDY

- The data collected and studied was limited to the Textile Sector in India. Other sectors in India may or may not react in the same manner to a stock-split.
- Within the Textile Sector, a few important companies were studied in detail, leaving out the other smaller companies that impact the sector.
- Data collected was only for the last five years. Trends may be different over longer periods of time.
- Fundamental and macro-economic factors were not taken into account while testing the hypotheses.

REFERENCES

- Bae, S.C. and Jo, H. (1999). "The Impact of Information Release on Stock Price Volatility and Trading Volume: The Rights Offering Case", *Review of Quantitative Finance and Accounting*, Vol.13, pp.153-169.
- Baker, H. Kent and Patricia, L. Gallagher (1980). "Managements" View of Stock Split," *Financial Management*, Vol.9, No.2, Summer, pp.73-77. [7].
- Brennan, Michael J., and Avanidhar Subrahmanyam (1996). "Market Microstructure and Asset Pricing: on the Compensation of Illiquidity in Stock Returns," *Journal of Financial Economics*, Vol.41, pp.441-446.
- Bris, A., W.N. Goetzmann and N. Zhu (2004). „Efficiency and the Bear: Short Sales and Markets around the World," *Yale School of Management Working Paper Series No.15*.
- Brown, D P and Z M Zhang (1997). "Market Orders and Market Efficiency," *Journal of Finance*," March, Vol. 52, No.1, pp.277-308 [21] .
- Compendium of Textile Statistics, Textile Commission of India, Bombay.
- Copeland, E. (1979). Liquidity changes following stock splits. *Journal of Finance* 34, 115-141.
- Fama, E. F., L. Fisher, M. Jensen and R. Roll (1969). "The Adjustment of Stock Prices to New Information," *International Economic Review*, Vol.10, pp.1-21.
- Froot, K., & Dabora, E. (1999): 'How are Stock Prices Affected by the Location of Trade?', *Journal of Financial Economics*, 53, 189- 216.

- Gupta S.C., Kapoor V.K. Mathematical Statistics.Probability. 201- 211.
- Kathuria, S. and A. Bhardwaj, (1998), "Export Quotas and Policy Constraints in the Indian Textile and Garment Industries," World Bank Policy Research Working Paper No. 2012, Washington DC, PP 47-50.
- Lamoureux, G., and Poon, P. (1987). The Market Reaction to Stock Splits. Journal of Finance, Vol. 42, No. 5, p. 1347-1370.
- Lin, J.C., Singh, A.K., Yu, W., 2009. Stock splits, trading continuity, and the cost of equity capital. Journal of Financial Economics 93, 474-489
- Marimuthu,K.N (2012),"Financial performance of textile industry :A study on listed companies of Tamil nadu" international journal of research in management, economic and commerce,vol.2 ,11 nov
- Pilotte, E. and T. Manuel, 1996, the market's response to recurring events. The case of stock splits, Journal of Financial Economics 41, 111-127.
- Reilly, F.K., Gustavson, S.G., 1985. Investing in options on stocks announcing splits. Financial Review 20, 121-142
- Schultz, P. (2000):\Stock Splits, Tick Size, and Sponsorship," The Journal of Finance, 55(1):429-450
- Shah.R.J .Descriptive Statistics. Testing of hypothesis, 65-69, Chi-Square distribution.89- 95.
- Vermaelen, T., 1981, Common stock repurchases and market signaling: An empirical study, Journal of Financial Economics 9, 139-183.
- Woolridge, R., 1983b, Stock dividends as signals, Journal of Financial Research 6, 1-12.
- Wulff, C. (2002): The Market Reaction to Stock Splits, Evidence from Germany," Schmalenbach Business Review, 54(3):280-290.